Magnetic Disk:

→ foundation of external memory.

Disk: circular platter
nonmagnetic material → Substrate → aluminium, glass → uniformity of magnetic film
magnetizable ∥ coating → surface defect ↓
→ stiffness ↑
→ shock & damage resistant

Data Organization:

Head → read & store data.
↳ data organized on the platter in a concentric set of rings → tracks
same width as head.

bit near center travels slower than bits on the outside.
Solution:
Defining variable spacing.
Outermost track has sectors with bigger spacing.
↳ give const. ang. velocity.

Bad Sector:

Permanently damaged storage unit.

Data Formatting:

- A possible way to locate a sector within the tracks is defining start and end points on each of the tracks and each of the sectors – handled by control data recorded on the disk - imposing a coordinating system on disk surface

Disk formatting (by extra data) accessed by disk drive, not by user.

Read & Write Mechanism:

Data recorded & retrieved via a conducting coil.

Write Mechanism: Electricity flowing through a coil produces a mg. field.
Electric pulses → wire head → magnetic pattern → recorded on the surface below.

Read Mechanism: A mg field moving relative to a coil produces an electrical current in the coil. (reverse)

When the surface of the disk rotates under the head, it generates a current of the same polarity as the one already recorded

head must generate an electromagnetic field

Narrow head → Narrow tracks → greater data density.

Head closer to disk → greater risk of error.

Disk performance parameter:

wait for device → wait for channel → seek → Rotational delay → Data transfer

seek → Track selection

Rotational delay → sector selection

Device busy

Disk Striping

Dividing a body of data into blocks & spreading it across multiple storage devices.

Standard Raid Levels

RAID 0

→ striping ✓ redundancy ✗

best performance fault tolerance ✗

RAID 1 (disk mirroring)

→ minimum 2 drives → duplicates data storage

striping ✗ read performance ↑

RAID 2

→ striping across disks

→ Stripping across disks
Error checking & correcting info (Hamming Code)

RAID 3
→ Striping ✓
1 drive stores parity information

RAID 4
→ large stripes (read records from single drive)

RAID 5
→ block level striping with parity
→ parity info striped across each drive.

RAID 6
→ similar to RAID 5
→ +2nd parity scheme (additional parity ⇒ array keeps functioning even if 2 disks fail together.

Nested RAID levels

RAID 10 (1+0) ⇒ Data is mirrored & the mirrors are striped.

RAID 01 (0+1): creates a stripe set which is mirrored.

SSD (Solid State Drive)

complement/Replacement of HDDs (both as internal & external memories)

Solid state → semiconductor built electronic ckt.

OS accessing SSD

On host system,

OS invokes file system software (File explorer) : access data on the disk → " " invoke i/o drivers software → i/o ds provides host access to particular SSD.

SSD internal → CI : PCIe
" external → " : USB
} CI → common Interface

SDD? slows down cuz

entire block read & placed in a RAM buffer

before writing back to flash memory, the entire block of flash Memory must be erased. Entire block from the buffer written back to FM.

FM unusable after certain number of writes.

Optical Memory

compact Disk: